

Assignment 1: Maritime "Shadow Fleet" Detection with Parallel Computing

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Low-Memory Streaming Partitioner

Streaming Pipeline — the CSV is never fully loaded into RAM:

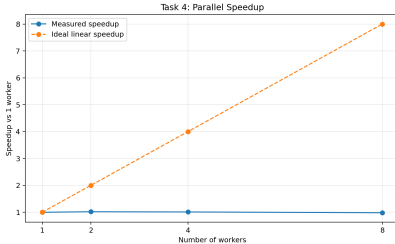
- 1 `csv.DictReader` reads **line-by-line** from disk
- 2 Generator `yield` passes one row at a time — $O(1)$ memory per row
- 3 `_chunked()` batches rows into **50 000-row** lists
- 4 `Pool.imap_unordered` dispatches chunks to **N worker processes** — results stream back as each worker finishes

The Dirty Data Trap — unconfigured transponders broadcast default MMSIs. Grouping naively $\Rightarrow$ mega-group with millions of rows $\Rightarrow$ worker OOM crash.

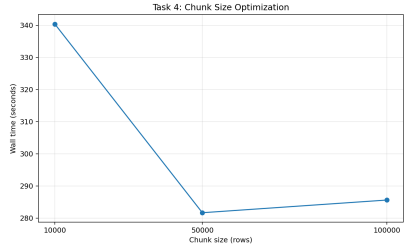
Multi-layer filter:

- 1 **Exact blacklist:** 000000000, 111111111
- 2 **Length check:** exactly 9-digits required
- 3 **Numeric range:** 100 000 000 - 999 999 999
- 4 **Non-numeric reject:** N/A, Unknown, empty
- 5 **Timestamp / coords:** invalid parsed out in processor

Speedup & Chunk Optimization



$S_8 \approx 1.0\times$ — pipeline is I/O-bound
Amdahl's parallel fraction $p \approx 0.15$



Optimal: **50 000** rows per chunk
10K $\rightarrow$ 50K saves $\sim 17\%$ wall time

Memory Profile



Metric

Peak RAM (total, all processes)

Assignment limit

Spike at ~150 s

Decay after ~800 s

Value

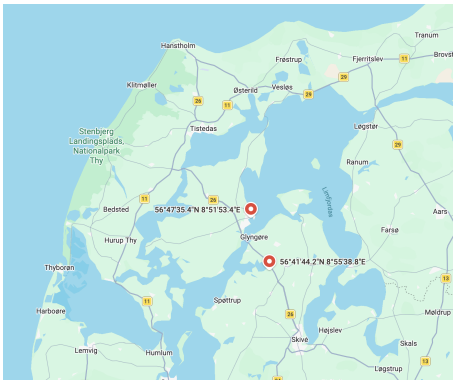
≈ 6 GB

< 1 GB/core × 8 cores = 8 GB

Reduce phase merging all timelines

GC as anomaly results are written

Real-World Verification — Top DFSI Vessel

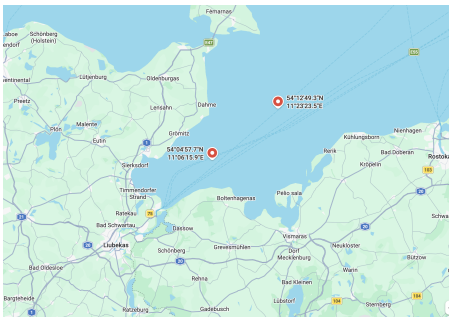


MMSI	219020897
Name	AN AIS
Type	Sailing vessel
Flag	Denmark
DFSI	235.85
D events	6
Peak speed	248 kn
Gap	90 seconds

Verdict: false positive.

Recreational sailboat in the Limfjorden, Denmark — not a shadow fleet zone. Class B AIS transponder with GPS glitches caused ~6 nm position jumps in 90 s intervals.

Real-World Verification — Going Dark



MMSI	266247000
Name	TUNDRALAND
Type	Ro-Ro Cargo, 190 m
Flag	Sweden
Cargo	Dangerous (IMO A)
DFSI	10.07 (rank #14)
Anomaly	A — Going Dark
Gap	20.1 hours
Distance	12.7 nm
Implied speed	0.63 kn

Thank you!